Table 1: Rayleigh–Bénard: spectral coherence RMSE per band (averaged over rollout horizons) and CRPS per rollout horizon, including the latent-space baseline LatentFM$_{\div 64}$ (autoencoder of [rozet2025]). Mean $\pm$ std over three inference seeds (42, 84, 168).

Model	Spectral coherence RMSE			CRPS		
	Low	Mid	High	01:32	33:64	65:196
WFM 1-scale	0.168 ± 0.005	$\underline{0.118 \pm 0.001}$	$\underline{0.077 \pm$ ¡0.001	$0.007 \pm$ ¡0.001	$0.041 \pm$ ¡0.001	$\underline{0.073 \pm 0.003}$
WFM 3-scale	$\mathbf{0.143 \pm 0.002}$	$\mathbf{0.109 \pm 0.001}$	$\mathbf{0.075 \pm 0.001}$	$\mathbf{0.004 \pm$ ¡0.001	$\mathbf{0.032 \pm$ ¡0.001	$\mathbf{0.069 \pm 0.001}$
$\mathrm{FM}_{\mathrm{pixel}}$	$\underline{0.149 \pm 0.001}$	$0.129 \pm$ ¡0.001	0.115 ± 0.001	$\underline{0.005 \pm$ ¡0.001	$\underline{0.036 \pm 0.001}$	0.076 ± 0.005
$\mathrm{FM}_{\mathrm{fourier}}$	0.294 ± 0.002	$0.332 \pm$ ¡0.001	$0.113 \pm$ ¡0.001	$0.013 \pm$ ¡0.001	0.040 ± 0.001	0.090 ± 0.001
WDNO	0.392 ± 0.001	$0.474 \pm$ ¡0.001	$0.535 \pm$ ¡0.001	$0.050 \pm$ ¡0.001	$0.102 \pm$ ¡0.001	$0.138 \pm$ ¡0.001
LatentFM$_{\div 64}$	$0.196 \pm$ ¡0.001	$0.159 \pm$ ¡0.001	0.131 ± 0.001	$0.026 \pm$ ¡0.001	$0.057 \pm$ ¡0.001	0.088 ± 0.004

Table 2: Kinetic-energy relative MAE across rollout horizons (error normalised by the trajectory time-mean of the true kinetic energy). Mean ± std over three inference seeds (42, 84, 168).

Rollout	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	01:32	33:64	65:97	01:32	33:64	65:196	01:16	17:32	33:77
WFM 1-scale	0.047 ± 0.001	0.113 ± 0.002	0.221 ± 0.006	0.007 ± ¡0.001	0.410 ± 0.005	0.379 ± 0.010	0.502 ± 0.001	0.538 ± 0.005	0.508 ± 0.001
WFM 3-scale	0.017 ± 0.001	**0.067 ± 0.003**	**0.147 ± 0.004**	**0.003 ± ¡0.001**	0.217 ± 0.006	0.348 ± 0.023	0.112 ± 0.002	0.391 ± 0.011	0.531 ± 0.003
FM_{pixel}	**0.016 ± 0.001**	**0.067 ± 0.004**	0.177 ± 0.007	0.004 ± ¡0.001	0.235 ± 0.005	**0.251 ± 0.010**	**0.051 ± 0.001**	**0.288 ± 0.012**	**0.320 ± 0.010**
$FM_{fourier}$	0.054 ± 0.001	0.096 ± 0.006	0.169 ± 0.011	0.015 ± ¡0.001	**0.157 ± 0.002**	0.919 ± 0.006	0.503 ± 0.002	0.538 ± 0.013	0.692 ± 0.008
WDNO	0.356 ± ¡0.001	0.225 ± 0.001	0.457 ± 0.002	1.014 ± 0.002	2.506 ± 0.007	2.309 ± 0.003	0.747 ± 0.002	0.890 ± 0.002	1.052 ± 0.002

Table 3: TRL, WFM 3-scale (haar): CRPS across rollout horizons under Gaussian perturbations of the initial conditions with standard deviation $\eta\,\sigma_c$. Mean $\pm$ std over three seeds.

IC noise	01:32	33:64	65:97
$\eta = 0$	0.410 ± 0.003	1.018 ± 0.011	1.178 ± 0.030
$\eta = 0.001$	0.410 ± 0.003	1.036 ± 0.016	1.180 ± 0.024
$\eta = 0.01$	0.410 ± 0.003	1.030 ± 0.016	1.190 ± 0.007
$\eta = 0.1$	0.413 ± 0.003	1.026 ± 0.040	1.203 ± 0.012

Table 4: TRL, WFM 3-scale (haar): Spectral coherence RMSE across bands under Gaussian perturbations of the initial conditions with standard deviation $\eta\,\sigma_c$. Mean $\pm$ std over three seeds.

IC noise	Low	Mid	High
$\eta = 0$	0.133 ± 0.001	$0.174 \pm$ ¡0.001	$0.141 \pm$ ¡0.001
$\eta = 0.001$	0.133 ± 0.001	0.174 ± 0.001	$0.141 \pm$ ¡0.001
$\eta = 0.01$	0.133 ± 0.001	0.174 ± 0.001	0.141 ± 0.001
$\eta = 0.1$	$0.133 \pm$ ¡0.001	$0.174 \pm$ ¡0.001	0.141 ± 0.001

Table 5: VRMSE across rollout horizons for all models and PDE systems. Deterministic baselines (WNO, FNO, TFNO) are single runs. Mean ± std over three inference seeds (42, 84, 168).

Rollout	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	01:32	33:64	65:97	01:32	33:64	65:196	01:16	17:32	33:77
WFM-1 haar	0.544 ± 0.011	0.791 ± 0.016	0.789 ± 0.014	1.782 ± 0.040	2.020 ± 0.049	0.788 ± 0.023	2.847 ± 0.015	1.541 ± 0.011	1.079 ± 0.004
WFM-1 db2	0.550 ± 0.003	0.750 ± 0.010	0.850 ± 0.024	1.333 ± 0.046	1.844 ± 0.103	0.791 ± 0.010	5.652 ± 0.019	2.930 ± 0.004	1.622 ± 0.003
WFM-1 db4	0.550 ± 0.007	0.730 ± 0.007	0.811 ± 0.001	1.689 ± 0.016	2.271 ± 0.077	0.802 ± 0.015	14.227 ± 0.027	5.574 ± 0.011	2.488 ± 0.001
WFM-1 db6	**0.536 ± 0.005**	**0.703 ± 0.009**	**0.786 ± 0.017**	1.441 ± 0.012	1.949 ± 0.028	1.025 ± 0.019	6.732 ± 0.009	2.927 ± 0.012	1.553 ± 0.010
WFM-3 haar	0.549 ± 0.007	0.824 ± 0.017	0.797 ± 0.014	1.559 ± 0.032	2.318 ± 0.095	0.807 ± 0.017	0.848 ± 0.008	1.327 ± 0.015	1.022 ± 0.002
WFM-3 db2	0.655 ± 0.006	0.893 ± 0.016	0.981 ± 0.022	1.051 ± 0.043	1.729 ± 0.048	**0.766 ± 0.008**	0.933 ± 0.011	1.349 ± 0.033	1.085 ± 0.007
WFM-3 db4	0.586 ± 0.015	0.759 ± 0.007	0.803 ± 0.006	1.029 ± 0.059	1.869 ± 0.048	0.770 ± 0.009	3.242 ± 0.013	2.271 ± 0.021	1.418 ± 0.005
WFM-3 db6	0.591 ± 0.005	0.879 ± 0.007	0.831 ± 0.019	**0.755 ± 0.025**	1.598 ± 0.059	0.834 ± 0.018	2.341 ± 0.010	2.118 ± 0.012	1.276 ± 0.006
FM_{pixel}	0.574 ± 0.002	0.891 ± 0.010	0.830 ± 0.016	1.319 ± 0.030	1.962 ± 0.030	0.821 ± 0.025	**0.580 ± 0.004**	1.380 ± 0.018	1.028 ± 0.002
$FM_{fourier}$	0.635 ± 0.010	0.773 ± 0.016	0.804 ± 0.007	2.331 ± 0.057	**1.417 ± 0.056**	0.870 ± 0.005	1.386 ± 0.003	**1.126 ± 0.008**	**1.007 ± 0.003**
WDNO	1.517 ± ¡0.001	1.834 ± 0.001	1.953 ± 0.003	19.262 ± 0.012	8.171 ± 0.032	1.392 ± 0.002	4.475 ± 0.011	2.181 ± 0.002	1.400 ± ¡0.001
WNO	inf	inf	–	44.226	24.463	2.452	10.669	4.321	2.002
FNO	2.952	4.978	4.349	3.368	3.972	inf	1.682	3.227	1.727
TFNO	2.300	6.534	6.387	102.678	31.842	2.965	4.443	2.968	1.525

Table 6: CRPS across rollout horizons for all models and PDE systems. Mean ± std over three inference seeds (42, 84, 168).

Rollout	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	01:32	33:64	65:97	01:32	33:64	65:196	01:16	17:32	33:77
WFM-1 haar	0.492 ± 0.006	1.047 ± 0.033	1.218 ± 0.032	0.005 ± ¡0.001	0.037 ± ¡0.001	0.074 ± 0.003	0.119 ± 0.001	0.190 ± 0.001	0.198 ± 0.001
WFM-1 db2	0.447 ± 0.007	1.097 ± 0.008	1.316 ± 0.032	0.005 ± ¡0.001	0.036 ± ¡0.001	0.073 ± 0.002	0.173 ± ¡0.001	0.239 ± ¡0.001	0.261 ± ¡0.001
WFM-1 db4	0.521 ± 0.006	1.052 ± 0.002	1.241 ± 0.021	0.007 ± ¡0.001	0.041 ± ¡0.001	0.073 ± 0.003	0.326 ± ¡0.001	0.388 ± ¡0.001	0.387 ± ¡0.001
WFM-1 db6	0.451 ± 0.003	**0.983 ± 0.017**	1.188 ± 0.008	0.005 ± ¡0.001	0.040 ± ¡0.001	0.090 ± 0.001	0.175 ± ¡0.001	0.222 ± ¡0.001	0.231 ± ¡0.001
WFM-3 haar	**0.410 ± 0.003**	1.018 ± 0.011	1.178 ± 0.030	0.004 ± ¡0.001	0.037 ± ¡0.001	0.077 ± 0.004	0.080 ± ¡0.001	0.169 ± 0.003	**0.191 ± 0.001**
WFM-3 db2	0.465 ± 0.008	1.133 ± 0.006	1.179 ± 0.026	**0.003 ± ¡0.001**	0.032 ± ¡0.001	**0.069 ± 0.002**	0.087 ± 0.001	0.176 ± 0.003	0.196 ± 0.002
WFM-3 db4	0.427 ± 0.004	0.984 ± 0.007	**1.130 ± 0.016**	0.004 ± ¡0.001	0.032 ± ¡0.001	**0.069 ± 0.001**	0.109 ± 0.001	0.209 ± 0.001	0.239 ± ¡0.001
WFM-3 db6	0.451 ± 0.006	1.026 ± 0.006	1.200 ± 0.015	0.004 ± ¡0.001	**0.031 ± 0.001**	0.079 ± 0.003	0.100 ± 0.001	0.197 ± 0.001	0.213 ± 0.001
FM_{pixel}	0.418 ± 0.006	1.088 ± 0.007	1.210 ± 0.017	0.005 ± ¡0.001	0.036 ± 0.001	0.076 ± 0.005	**0.059 ± ¡0.001**	**0.166 ± 0.001**	0.194 ± 0.001
$FM_{fourier}$	1.112 ± 0.003	1.632 ± 0.007	1.772 ± 0.011	0.013 ± ¡0.001	0.040 ± 0.001	0.090 ± 0.001	0.197 ± 0.001	0.216 ± 0.002	0.211 ± 0.001
WDNO	2.460 ± 0.002	3.374 ± 0.004	4.284 ± 0.013	0.050 ± ¡0.001	0.102 ± ¡0.001	0.138 ± ¡0.001	0.218 ± ¡0.001	0.245 ± ¡0.001	0.266 ± ¡0.001

Table 7: Spectral coherence RMSE across *Low*, *Mid*, and *High* frequency bands for all models and PDE systems, averaged over rollout horizons. Mean ± std over three inference seeds (42, 84, 168).

Frequency Bands	Turb. Rad. Layer 2D			Rayleigh–Bénard			Active Matter		
	Low	Mid	High	Low	Mid	High	Low	Mid	High
WFM-1 haar	**0.132 ± ¡0.001**	0.175 ± 0.001	**0.134 ± ¡0.001**	0.171 ± 0.002	0.132 ± ¡0.001	0.116 ± 0.001	0.229 ± 0.002	0.291 ± ¡0.001	0.204 ± ¡0.001
WFM-1 db2	0.139 ± ¡0.001	0.181 ± ¡0.001	0.142 ± ¡0.001	0.167 ± 0.001	0.126 ± 0.001	0.105 ± 0.001	0.291 ± ¡0.001	0.336 ± ¡0.001	0.236 ± ¡0.001
WFM-1 db4	**0.132 ± ¡0.001**	0.175 ± ¡0.001	0.140 ± ¡0.001	0.168 ± 0.005	0.118 ± 0.001	0.077 ± ¡0.001	0.306 ± 0.001	0.351 ± ¡0.001	0.251 ± ¡0.001
WFM-1 db6	0.133 ± 0.001	0.175 ± ¡0.001	0.141 ± ¡0.001	0.196 ± 0.003	0.131 ± 0.001	0.093 ± 0.002	0.283 ± 0.001	0.346 ± ¡0.001	0.242 ± ¡0.001
WFM-3 haar	0.133 ± 0.001	**0.174 ± ¡0.001**	0.141 ± ¡0.001	0.149 ± 0.001	0.130 ± 0.001	0.114 ± 0.001	0.205 ± 0.001	0.280 ± 0.001	0.256 ± ¡0.001
WFM-3 db2	0.139 ± 0.002	0.188 ± 0.001	0.136 ± ¡0.001	**0.141 ± 0.003**	0.113 ± 0.001	0.079 ± 0.001	0.199 ± 0.001	0.262 ± ¡0.001	0.198 ± ¡0.001
WFM-3 db4	0.141 ± 0.001	0.178 ± ¡0.001	0.144 ± ¡0.001	0.143 ± 0.002	**0.109 ± 0.001**	**0.075 ± 0.001**	0.219 ± 0.001	0.292 ± ¡0.001	0.235 ± ¡0.001
WFM-3 db6	0.136 ± ¡0.001	0.175 ± 0.001	0.148 ± ¡0.001	0.147 ± 0.001	0.110 ± ¡0.001	0.080 ± 0.001	0.196 ± 0.001	**0.258 ± ¡0.001**	0.260 ± ¡0.001
FM_{pixel}	0.141 ± 0.001	0.181 ± 0.001	0.138 ± 0.001	0.149 ± 0.001	0.129 ± ¡0.001	0.115 ± 0.001	**0.183 ± 0.002**	**0.258 ± ¡0.001**	0.291 ± ¡0.001
$FM_{fourier}$	0.240 ± 0.002	0.686 ± ¡0.001	0.236 ± ¡0.001	0.294 ± 0.002	0.332 ± ¡0.001	0.113 ± ¡0.001	0.436 ± 0.001	0.534 ± ¡0.001	**0.001 ± ¡0.001**
WDNO	0.199 ± ¡0.001	0.249 ± ¡0.001	0.366 ± ¡0.001	0.392 ± 0.001	0.474 ± ¡0.001	0.535 ± ¡0.001	0.367 ± 0.001	0.345 ± ¡0.001	0.200 ± ¡0.001

Table 8: CRPS per t_{cool} on TRL, averaged over rollout steps. Mean $\pm$ std over three inference seeds (42, 84, 168).

t_{cool}	0.03	0.06	0.1	0.18	0.32	0.56	1	1.78	3.16
WFM-1 haar	1.165 ± 0.043	1.135 ± 0.057	0.865 ± 0.052	1.044 ± 0.028	0.830 ± 0.051	0.798 ± 0.030	0.806 ± 0.056	0.983 ± 0.036	0.673 ± 0.023
WFM-1 db2	1.112 ± 0.031	1.078 ± 0.025	0.878 ± 0.042	1.057 ± 0.040	0.880 ± 0.065	0.779 ± 0.015	0.853 ± 0.053	0.976 ± 0.051	1.002 ± 0.061
WFM-1 db4	1.170 ± 0.048	1.102 ± 0.044	0.847 ± 0.043	1.084 ± 0.052	0.875 ± 0.005	0.849 ± 0.019	0.879 ± 0.065	0.946 ± 0.038	0.719 ± 0.036
WFM-1 db6	1.076 ± 0.023	1.074 ± 0.014	$\underline{0.831 \pm 0.021}$	1.041 ± 0.017	0.823 ± 0.021	0.773 ± 0.034	0.741 ± 0.016	$\underline{0.943 \pm 0.018}$	$\underline{0.594 \pm 0.040}$
WFM-3 haar	1.107 ± 0.012	1.117 ± 0.034	0.868 ± 0.019	$\underline{0.988 \pm 0.067}$	$\mathbf{0.773 \pm 0.020}$	$\underline{0.765 \pm 0.046}$	$\mathbf{0.675 \pm 0.026}$	0.961 ± 0.011	0.595 ± 0.024
WFM-3 db2	1.248 ± 0.024	1.141 ± 0.044	0.894 ± 0.043	1.027 ± 0.025	0.836 ± 0.017	0.783 ± 0.041	0.751 ± 0.040	1.019 ± 0.008	0.656 ± 0.028
WFM-3 db4	$\mathbf{1.048 \pm 0.019}$	$\mathbf{1.006 \pm 0.025}$	$\mathbf{0.817 \pm 0.042}$	$\mathbf{0.982 \pm 0.049}$	$\underline{0.817 \pm 0.033}$	$\mathbf{0.749 \pm 0.027}$	$\underline{0.692 \pm 0.028}$	$\mathbf{0.906 \pm 0.005}$	0.632 ± 0.011
WFM-3 db6	$\underline{1.075 \pm 0.018}$	$\underline{1.034 \pm 0.015}$	0.882 ± 0.038	1.000 ± 0.053	0.863 ± 0.039	0.850 ± 0.027	0.770 ± 0.048	0.969 ± 0.038	0.617 ± 0.006
FM_{pixel}	1.220 ± 0.040	1.126 ± 0.016	0.875 ± 0.045	1.061 ± 0.048	0.855 ± 0.046	0.783 ± 0.004	0.731 ± 0.012	0.978 ± 0.021	$\mathbf{0.551 \pm 0.030}$
$\text{FM}_{\text{fourier}}$	1.709 ± 0.031	1.667 ± 0.032	1.424 ± 0.006	1.618 ± 0.012	1.485 ± 0.019	1.430 ± 0.016	1.451 ± 0.014	1.595 ± 0.015	1.194 ± 0.003
WDNO	3.330 ± 0.008	3.292 ± 0.007	3.183 ± 0.005	3.419 ± 0.005	3.369 ± 0.011	3.409 ± 0.010	3.459 ± 0.010	3.552 ± 0.012	3.424 ± 0.011

Table 9: Spectral coherence RMSE per t_{cool} and frequency band on TRL. Mean ± std over three inference seeds (42, 84, 168).

Model	t_{cool}	0.03	0.06	0.1	0.18	0.32	0.56	1	1.78	3.16
WFM-1 haar	Low	0.138 ± 0.007	**0.139 ± 0.003**	0.138 ± 0.003	0.138 ± 0.004	**0.132 ± 0.001**	0.125 ± 0.004	0.127 ± 0.005	0.136 ± 0.003	0.121 ± 0.005
	Mid	0.190 ± 0.001	0.194 ± 0.001	0.189 ± 0.001	0.189 ± 0.001	**0.180 ± 0.001**	0.172 ± 0.002	0.157 ± 0.004	0.155 ± 0.001	0.150 ± 0.004
	High	0.138 ± 0.001	**0.138 ± 0.001**	**0.136 ± 0.002**	**0.140 ± 0.001**	0.147 ± 0.001	0.147 ± 0.001	0.133 ± 0.001	0.120 ± 0.001	0.106 ± 0.001
WFM-1 db2	Low	0.141 ± 0.002	**0.139 ± 0.001**	0.140 ± 0.004	0.137 ± 0.003	0.143 ± 0.005	0.134 ± 0.003	0.136 ± 0.003	0.134 ± 0.001	0.149 ± 0.002
	Mid	0.192 ± 0.002	0.195 ± 0.001	0.189 ± 0.004	0.193 ± 0.002	0.189 ± 0.004	0.179 ± 0.001	0.170 ± 0.002	0.163 ± 0.001	0.164 ± 0.002
	High	0.142 ± 0.002	0.143 ± 0.001	0.139 ± 0.003	0.145 ± 0.001	0.153 ± 0.002	0.155 ± 0.001	0.145 ± 0.001	0.134 ± 0.001	0.122 ± 0.001
WFM-1 db4	Low	**0.136 ± 0.003**	0.141 ± 0.002	**0.134 ± 0.002**	**0.134 ± 0.003**	0.136 ± 0.001	0.130 ± 0.002	0.127 ± 0.002	**0.129 ± 0.002**	0.123 ± 0.002
	Mid	**0.189 ± 0.002**	**0.192 ± 0.001**	**0.185 ± 0.002**	**0.188 ± 0.002**	0.184 ± 0.001	0.177 ± 0.001	0.162 ± 0.001	0.154 ± 0.001	0.148 ± 0.001
	High	**0.136 ± 0.001**	**0.138 ± 0.001**	0.137 ± 0.001	0.144 ± 0.001	0.151 ± 0.001	0.158 ± 0.001	0.147 ± 0.001	0.135 ± 0.001	0.112 ± 0.001
WFM-1 db6	Low	0.138 ± 0.001	0.143 ± 0.005	0.139 ± 0.005	0.136 ± 0.002	0.136 ± 0.001	**0.124 ± 0.002**	0.128 ± 0.001	0.134 ± 0.001	0.120 ± 0.005
	Mid	0.190 ± 0.001	0.196 ± 0.002	0.191 ± 0.003	0.191 ± 0.001	0.183 ± 0.001	0.172 ± 0.001	0.158 ± 0.003	0.151 ± 0.001	0.146 ± 0.002
	High	0.139 ± 0.001	0.140 ± 0.001	0.141 ± 0.001	0.147 ± 0.001	0.154 ± 0.001	0.160 ± 0.001	0.145 ± 0.001	0.131 ± 0.001	0.112 ± 0.001
WFM-3 haar	Low	0.143 ± 0.001	0.144 ± 0.005	0.139 ± 0.003	0.136 ± 0.004	0.133 ± 0.005	0.127 ± 0.001	**0.123 ± 0.003**	0.132 ± 0.001	**0.118 ± 0.003**
	Mid	0.194 ± 0.001	0.199 ± 0.002	0.188 ± 0.001	0.190 ± 0.001	0.181 ± 0.003	**0.167 ± 0.001**	**0.153 ± 0.001**	**0.150 ± 0.001**	**0.144 ± 0.001**
	High	0.152 ± 0.002	0.152 ± 0.002	0.148 ± 0.002	0.148 ± 0.001	0.157 ± 0.001	0.152 ± 0.001	0.135 ± 0.001	0.123 ± 0.001	0.103 ± 0.001
WFM-3 db2	Low	0.150 ± 0.002	0.146 ± 0.002	0.145 ± 0.003	0.148 ± 0.001	0.141 ± 0.007	0.136 ± 0.004	**0.123 ± 0.003**	0.141 ± 0.002	0.123 ± 0.003
	Mid	0.193 ± 0.001	0.196 ± 0.001	0.191 ± 0.001	0.201 ± 0.002	0.195 ± 0.005	0.188 ± 0.002	0.174 ± 0.002	0.180 ± 0.003	0.175 ± 0.001
	High	0.152 ± 0.002	0.151 ± 0.002	0.146 ± 0.002	0.143 ± 0.001	**0.141 ± 0.002**	**0.140 ± 0.001**	0.127 ± 0.001	0.120 ± 0.001	0.105 ± 0.001
WFM-3 db4	Low	0.151 ± 0.003	0.146 ± 0.003	0.141 ± 0.004	0.144 ± 0.004	0.146 ± 0.004	0.137 ± 0.004	0.132 ± 0.005	0.140 ± 0.002	0.131 ± 0.005
	Mid	0.195 ± 0.001	0.197 ± 0.002	0.189 ± 0.003	0.192 ± 0.002	0.189 ± 0.001	0.176 ± 0.003	0.162 ± 0.001	0.155 ± 0.001	0.147 ± 0.001
	High	0.149 ± 0.001	0.152 ± 0.001	0.150 ± 0.002	0.154 ± 0.002	0.159 ± 0.002	0.157 ± 0.001	0.141 ± 0.001	0.129 ± 0.001	0.109 ± 0.001
WFM-3 db6	Low	0.145 ± 0.002	0.147 ± 0.005	0.144 ± 0.003	0.139 ± 0.006	0.141 ± 0.002	0.125 ± 0.002	0.132 ± 0.003	0.136 ± 0.005	0.119 ± 0.001
	Mid	0.193 ± 0.001	0.198 ± 0.002	0.188 ± 0.002	0.191 ± 0.003	0.181 ± 0.002	0.168 ± 0.001	0.158 ± 0.003	0.152 ± 0.003	0.147 ± 0.003
	High	0.148 ± 0.001	0.150 ± 0.003	0.151 ± 0.001	0.156 ± 0.003	0.170 ± 0.001	0.164 ± 0.001	0.144 ± 0.001	0.134 ± 0.001	0.112 ± 0.001
FM_{pixel}	Low	0.144 ± 0.004	0.142 ± 0.003	0.142 ± 0.002	0.143 ± 0.003	0.141 ± 0.003	0.143 ± 0.003	0.137 ± 0.004	0.154 ± 0.004	0.126 ± 0.005
	Mid	0.193 ± 0.002	0.199 ± 0.002	0.191 ± 0.001	0.196 ± 0.002	0.186 ± 0.003	0.187 ± 0.001	0.169 ± 0.004	0.163 ± 0.004	0.151 ± 0.003
	High	0.153 ± 0.004	0.157 ± 0.001	0.149 ± 0.002	0.150 ± 0.001	0.153 ± 0.002	0.142 ± 0.001	**0.126 ± 0.001**	**0.113 ± 0.001**	**0.098 ± 0.001**
$FM_{fourier}$	Low	0.252 ± 0.003	0.250 ± 0.003	0.238 ± 0.002	0.255 ± 0.004	0.240 ± 0.002	0.232 ± 0.003	0.235 ± 0.002	0.244 ± 0.003	0.214 ± 0.001
	Mid	0.680 ± 0.001	0.674 ± 0.002	0.677 ± 0.001	0.682 ± 0.001	0.686 ± 0.001	0.689 ± 0.001	0.694 ± 0.001	0.688 ± 0.001	0.705 ± 0.001
	High	0.224 ± 0.001	0.223 ± 0.001	0.224 ± 0.001	0.229 ± 0.001	0.233 ± 0.001	0.238 ± 0.001	0.244 ± 0.001	0.249 ± 0.001	0.258 ± 0.001
WDNO	Low	0.204 ± 0.001	0.203 ± 0.001	0.194 ± 0.001	0.205 ± 0.001	0.195 ± 0.001	0.192 ± 0.001	0.195 ± 0.001	0.215 ± 0.001	0.190 ± 0.002
	Mid	0.253 ± 0.001	0.252 ± 0.001	0.250 ± 0.001	0.246 ± 0.001	0.241 ± 0.001	0.244 ± 0.001	0.247 ± 0.001	0.251 ± 0.001	0.262 ± 0.001
	High	0.359 ± 0.001	0.363 ± 0.001	0.355 ± 0.001	0.363 ± 0.001	0.367 ± 0.001	0.370 ± 0.001	0.376 ± 0.001	0.378 ± 0.001	0.373 ± 0.001

Table 10: CRPS per Rayleigh number Ra on Rayleigh–Bénard, averaged over Prandtl numbers. Mean ± std over three inference seeds (42, 84, 168).

Ra	10^6	10^8	10^{10}
WFM-1 haar	0.059 ± 0.001	0.060 ± 0.005	0.051 ± 0.001
WFM-1 db2	$\mathbf{0.052 \pm 0.002}$	0.062 ± 0.003	0.052 ± 0.003
WFM-1 db4	0.059 ± 0.008	0.060 ± 0.003	0.051 ± 0.001
WFM-1 db6	0.060 ± 0.004	0.069 ± 0.004	0.075 ± 0.001
WFM-3 haar	0.060 ± 0.004	0.060 ± 0.003	0.055 ± 0.003
WFM-3 db2	0.054 ± 0.004	$\mathbf{0.055 \pm 0.001}$	$\underline{0.049 \pm 0.001}$
WFM-3 db4	$\underline{0.053 \pm 0.005}$	$\underline{0.058 \pm 0.002}$	$\mathbf{0.048 \pm 0.003}$
WFM-3 db6	0.065 ± 0.006	$\mathbf{0.055 \pm 0.002}$	0.057 ± 0.001
$\mathrm{FM}_{\mathrm{pixel}}$	0.057 ± 0.007	0.066 ± 0.004	0.051 ± 0.002
$\mathrm{FM}_{\mathrm{fourier}}$	0.069 ± 0.001	0.076 ± 0.001	0.062 ± 0.001
WDNO	0.120 ± 0.001	$0.117 \pm ¡0.001$	0.116 ± 0.001

Table 11: CRPS per Prandtl number Pr on Rayleigh–Bénard, averaged over Rayleigh numbers. Mean ± std over three inference seeds (42, 84, 168).

Pr	10^{-1}	1	10^{1}
WFM-1 haar	0.091 ± 0.001	0.050 ± 0.004	<u>0.029 ± 0.001</u>
WFM-1 db2	0.088 ± 0.005	<u>0.049 ± 0.002</u>	0.030 ± 0.001
WFM-1 db4	0.088 ± 0.007	**0.048 ± 0.002**	0.034 ± 0.001
WFM-1 db6	0.087 ± 0.003	0.066 ± 0.004	0.051 ± 0.001
WFM-3 haar	0.092 ± 0.007	0.050 ± 0.002	0.033 ± ¡0.001
WFM-3 db2	**0.079 ± 0.003**	**0.048 ± 0.001**	0.031 ± 0.001
WFM-3 db4	**0.079 ± 0.005**	0.051 ± 0.004	**0.028 ± 0.001**
WFM-3 db6	<u>0.086 ± 0.007</u>	0.050 ± 0.002	0.040 ± 0.001
$\mathrm{FM}_{\mathrm{pixel}}$	0.088 ± 0.008	0.054 ± 0.001	0.033 ± 0.001
$\mathrm{FM}_{\mathrm{fourier}}$	0.111 ± 0.001	0.058 ± 0.001	0.038 ± 0.001
WDNO	0.146 ± 0.001	0.112 ± ¡0.001	0.096 ± ¡0.001

Table 12: Spectral coherence RMSE per frequency band on Rayleigh–Bénard vs Ra (averaged over Pr). Mean ± std over three inference seeds (42, 84, 168).

Model	Ra	10^6	10^8	10^{10}
WFM-1 haar	Low	0.174 ± 0.002	0.166 ± 0.008	0.138 ± 0.003
	Mid	0.189 ± 0.002	0.135 ± 0.001	$0.133 \pm$ ¡0.001
	High	0.110 ± 0.002	0.186 ± 0.005	0.163 ± 0.001
WFM-1 db2	Low	0.154 ± 0.008	0.176 ± 0.008	0.140 ± 0.001
	Mid	0.161 ± 0.002	0.132 ± 0.001	$0.134 \pm$ ¡0.001
	High	0.092 ± 0.001	0.163 ± 0.004	0.150 ± 0.001
WFM-1 db4	Low	0.160 ± 0.021	0.169 ± 0.006	0.140 ± 0.003
	Mid	$\underline{0.121 \pm 0.006}$	0.130 ± 0.001	$0.131 \pm$ ¡0.001
	High	$\mathbf{0.049 \pm 0.002}$	0.102 ± 0.001	$\mathbf{0.120 \pm 0.001}$
WFM-1 db6	Low	0.199 ± 0.002	0.219 ± 0.008	0.210 ± 0.009
	Mid	0.160 ± 0.001	0.138 ± 0.004	0.178 ± 0.004
	High	0.066 ± 0.003	0.121 ± 0.006	0.183 ± 0.005
WFM-3 haar	Low	0.154 ± 0.005	0.163 ± 0.005	$\underline{0.130 \pm 0.002}$
	Mid	0.188 ± 0.004	0.133 ± 0.001	$0.129 \pm$ ¡0.001
	High	0.110 ± 0.003	0.181 ± 0.002	0.156 ± 0.002
WFM-3 db2	Low	$\underline{0.138 \pm 0.003}$	$\underline{0.159 \pm 0.013}$	$\mathbf{0.127 \pm 0.001}$
	Mid	0.130 ± 0.004	$\underline{0.120 \pm 0.001}$	$\mathbf{0.125 \pm}$ **¡0.001**
	High	0.060 ± 0.002	0.104 ± 0.002	$\mathbf{0.120 \pm}$ **¡0.001**
WFM-3 db4	Low	$\mathbf{0.137 \pm 0.010}$	0.160 ± 0.006	$\underline{0.130 \pm 0.004}$
	Mid	$\mathbf{0.118 \pm 0.005}$	$\mathbf{0.119 \pm 0.001}$	$\mathbf{0.125 \pm}$ **¡0.001**
	High	$\underline{0.052 \pm 0.002}$	$\mathbf{0.097 \pm 0.003}$	$\mathbf{0.120 \pm 0.001}$
WFM-3 db6	Low	0.145 ± 0.007	0.166 ± 0.002	0.138 ± 0.002
	Mid	0.122 ± 0.002	$\underline{0.120 \pm}$ ¡0.001	0.130 ± 0.001
	High	0.066 ± 0.002	$\underline{0.099 \pm 0.001}$	$\underline{0.127 \pm 0.001}$
$\mathrm{FM}_{\mathrm{pixel}}$	Low	0.155 ± 0.002	$\mathbf{0.150 \pm 0.002}$	$0.131 \pm$ ¡0.001
	Mid	0.188 ± 0.002	0.130 ± 0.002	$\underline{0.128 \pm}$ ¡0.001
	High	0.112 ± 0.001	0.183 ± 0.006	0.156 ± 0.003
$\mathrm{FM}_{\mathrm{fourier}}$	Low	0.314 ± 0.002	0.319 ± 0.004	0.298 ± 0.002
	Mid	0.317 ± 0.001	$0.388 \pm$ ¡0.001	$0.398 \pm$ ¡0.001
	High	$0.110 \pm$ ¡0.001	$0.125 \pm$ ¡0.001	$\underline{0.127 \pm}$ ¡0.001
WDNO	Low	0.426 ± 0.003	0.402 ± 0.001	0.363 ± 0.001
	Mid	$0.549 \pm$ ¡0.001	$0.484 \pm$ ¡0.001	0.448 ± 0.001

Table 13: Spectral coherence RMSE per frequency band on Rayleigh–Bénard vs Pr (averaged over Ra). Mean ± std over three inference seeds (42, 84, 168).

Model	Pr	10^{-1}	1	10^{1}
WFM-1 haar	Low	0.187 ± 0.007	0.144 ± 0.003	0.147 ± 0.002
	Mid	0.158 ± 0.002	$0.164 \pm$ ¡0.001	$0.136 \pm$ ¡0.001
	High	0.192 ± 0.004	0.155 ± 0.001	0.111 ± 0.002
WFM-1 db2	Low	0.180 ± 0.011	0.145 ± 0.008	0.145 ± 0.002
	Mid	0.150 ± 0.003	0.153 ± 0.001	0.123 ± 0.002
	High	0.182 ± 0.001	0.136 ± 0.004	0.087 ± 0.001
WFM-1 db4	Low	0.181 ± 0.026	0.141 ± 0.006	0.148 ± 0.003
	Mid	0.138 ± 0.004	0.133 ± 0.002	$\underline{0.111 \pm \text{¡0.001}}$
	High	$\underline{0.120 \pm 0.001}$	$\mathbf{0.087 \pm 0.002}$	$\mathbf{0.063 \pm 0.001}$
WFM-1 db6	Low	0.200 ± 0.013	0.210 ± 0.010	0.217 ± 0.007
	Mid	0.148 ± 0.005	0.162 ± 0.004	0.165 ± 0.002
	High	0.129 ± 0.006	0.150 ± 0.006	0.091 ± 0.004
WFM-3 haar	Low	0.171 ± 0.001	0.139 ± 0.003	0.137 ± 0.001
	Mid	0.155 ± 0.003	0.159 ± 0.002	0.136 ± 0.002
	High	0.189 ± 0.004	0.150 ± 0.004	0.108 ± 0.002
WFM-3 db2	Low	$\mathbf{0.160 \pm 0.014}$	0.133 ± 0.006	$\mathbf{0.131 \pm 0.001}$
	Mid	$\underline{0.128 \pm 0.002}$	0.134 ± 0.001	0.112 ± 0.002
	High	0.122 ± 0.002	0.096 ± 0.002	$\underline{0.066 \pm 0.001}$
WFM-3 db4	Low	$\underline{0.163 \pm 0.006}$	$\mathbf{0.130 \pm 0.004}$	$\underline{0.134 \pm \text{¡0.001}}$
	Mid	$\mathbf{0.126 \pm 0.001}$	$\mathbf{0.128 \pm 0.002}$	$\mathbf{0.108 \pm 0.001}$
	High	$\mathbf{0.112 \pm 0.002}$	$\underline{0.090 \pm 0.001}$	$\underline{0.066 \pm \text{¡0.001}}$
WFM-3 db6	Low	0.172 ± 0.001	0.137 ± 0.003	$0.139 \pm$ ¡0.001
	Mid	0.129 ± 0.001	$\underline{0.132 \pm 0.001}$	$\underline{0.111 \pm \text{¡0.001}}$
	High	$\underline{0.120 \pm 0.003}$	0.094 ± 0.001	$0.079 \pm$ ¡0.001
FM_{pixel}	Low	0.169 ± 0.004	$\underline{0.132 \pm 0.002}$	0.136 ± 0.002
	Mid	0.153 ± 0.001	0.160 ± 0.002	0.133 ± 0.001
	High	0.197 ± 0.001	0.149 ± 0.007	0.106 ± 0.001
$\text{FM}_{\text{fourier}}$	Low	0.334 ± 0.004	0.307 ± 0.001	0.290 ± 0.004
	Mid	0.384 ± 0.001	$0.365 \pm$ ¡0.001	0.354 ± 0.001
	High	$0.123 \pm$ ¡0.001	$0.123 \pm$ ¡0.001	$0.116 \pm$ ¡0.001
WDNO	Low	0.390 ± 0.003	0.388 ± 0.003	0.413 ± 0.001
	Mid	$0.477 \pm$ ¡0.001	$0.498 \pm$ ¡0.001	0.505 ± 0.001

Table 14: Spectral Coherence RMSE averaged across *Low*, *Mid* and *High* frequency bands and rollout horizons per PDE system. For each WFM variant, **bold** denotes the optimal wavelet representation used in all subsequent experiments. Mean ± std over three inference seeds (42, 84, 168).

	Turb. Rad. Layer 2D	Rayleigh–Bénard	Active Matter
WFM-1 haar	**0.147 ± ¡0.001**	0.140 ± 0.001	**0.241 ± 0.001**
WFM-1 db2	0.154 ± ¡0.001	0.132 ± 0.001	0.288 ± ¡0.001
WFM-1 db4	0.149 ± ¡0.001	**0.121 ± 0.002**	0.303 ± ¡0.001
WFM-1 db6	0.150 ± ¡0.001	0.140 ± 0.002	0.290 ± ¡0.001
WFM-3 haar	**0.149 ± ¡0.001**	0.131 ± 0.001	0.247 ± 0.001
WFM-3 db2	0.154 ± 0.001	0.111 ± 0.001	**0.220 ± ¡0.001**
WFM-3 db4	0.154 ± ¡0.001	**0.109 ± 0.001**	0.249 ± ¡0.001
WFM-3 db6	0.153 ± 0.001	0.112 ± ¡0.001	0.238 ± ¡0.001
$\mathrm{FM}_{\mathrm{pixel}}$	0.153 ± 0.001	0.131 ± 0.001	0.244 ± 0.001
$\mathrm{FM}_{\mathrm{fourier}}$	0.387 ± 0.001	0.246 ± 0.001	0.324 ± ¡0.001
WDNO	0.271 ± ¡0.001	0.467 ± ¡0.001	0.304 ± ¡0.001

PDE	**WFM 1-scale**	**WFM 3-scale**
TRL	1.25×	1.14×
RB	1.52×	1.38×
AM	1.46×	1.26×

Table 15: CRPS per Rayleigh number Ra on Rayleigh–Bénard, averaged over Prandtl numbers. Mean ± std over three inference seeds (42, 84, 168).

Ra	10^6	10^8	10^{10}
WFM-1 haar	0.059 ± 0.001	0.060 ± 0.005	0.051 ± 0.001
WFM-1 db2	$\mathbf{0.052 \pm 0.002}$	0.062 ± 0.003	0.052 ± 0.003
WFM-1 db4	0.059 ± 0.008	0.060 ± 0.003	0.051 ± 0.001
WFM-1 db6	0.060 ± 0.004	0.069 ± 0.004	0.075 ± 0.001
WFM-3 haar	0.060 ± 0.004	0.060 ± 0.003	0.055 ± 0.003
WFM-3 db2	0.054 ± 0.004	$\mathbf{0.055 \pm 0.001}$	$\underline{0.049 \pm 0.001}$
WFM-3 db4	$\underline{0.053 \pm 0.005}$	$\underline{0.058 \pm 0.002}$	$\mathbf{0.048 \pm 0.003}$
WFM-3 db6	0.065 ± 0.006	$\mathbf{0.055 \pm 0.002}$	0.057 ± 0.001
$\mathrm{FM}_{\mathrm{pixel}}$	0.057 ± 0.007	0.066 ± 0.004	0.051 ± 0.002
$\mathrm{FM}_{\mathrm{fourier}}$	0.069 ± 0.001	0.076 ± 0.001	0.062 ± 0.001
WDNO	0.120 ± 0.001	$0.117 \pm ¡0.001$	0.116 ± 0.001

Table 16: CRPS per t_{cool} on TRL, averaged over rollout steps. Mean $\pm$ std over three inference seeds (42, 84, 168).

t_{cool}	0.03	0.06	0.1
WFM-1 haar	1.165 ± 0.043	1.135 ± 0.057	0.865 ± 0.052
WFM-1 db2	1.112 ± 0.031	1.078 ± 0.025	0.878 ± 0.042
WFM-1 db4	1.170 ± 0.048	1.102 ± 0.044	0.847 ± 0.043
WFM-1 db6	1.076 ± 0.023	1.074 ± 0.014	$\underline{0.831 \pm 0.021}$
WFM-3 haar	1.107 ± 0.012	1.117 ± 0.034	0.868 ± 0.019
WFM-3 db2	1.248 ± 0.024	1.141 ± 0.044	0.894 ± 0.043
WFM-3 db4	$\mathbf{1.048 \pm 0.019}$	$\mathbf{1.006 \pm 0.025}$	$\mathbf{0.817 \pm 0.042}$
WFM-3 db6	$\underline{1.075 \pm 0.018}$	$\underline{1.034 \pm 0.015}$	0.882 ± 0.038
FM_{pixel}	1.220 ± 0.040	1.126 ± 0.016	0.875 ± 0.045
$\text{FM}_{\text{fourier}}$	1.709 ± 0.031	1.667 ± 0.032	1.424 ± 0.006
WDNO	3.330 ± 0.008	3.292 ± 0.007	3.183 ± 0.005